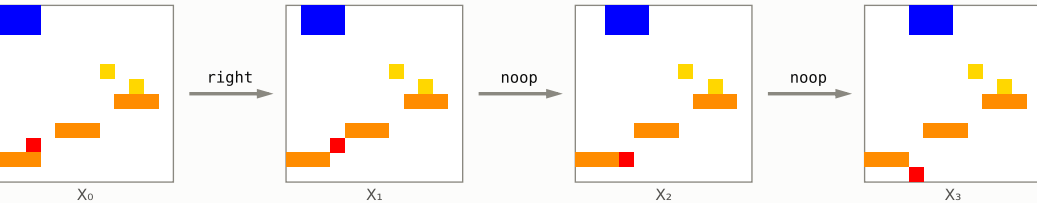


Mario (N2NTD) — three transitions from one training drive



$X_0 \rightarrow X_1$ right the player steps right, $(9,2) \rightarrow (9,3)$, off the end of the platform
 $X_1 \rightarrow X_2$ noop no action, and it falls one row
 $X_2 \rightarrow X_3$ noop and falls again, to the floor at row 11
on every one of these steps the blue 2×3 block also advances one column, whatever the agent does

	abstraction program P	world knowledge K
node 0 seed 0.00 min(ID, cFD)	<pre>blank slate — parses the grid, emits nothing grid = _parse_grid(obs) ... return "" P(X1) = (empty string) P(X2): identical — the transition is invisible to P</pre>	(empty — no world knowledge yet)
node 2 iter 3 · P 0.23 min(ID, cFD)	<pre>updated — one flat list of non-background cells parts_alt = [f"grid {rows}x{cols}; bg:{bg}; cells:"] for r, c, clr in cells: parts_alt.append(f"({r},{c},{clr});") summary = "".join(parts_alt) P(X1) = grid 12x12; bg:white; cells:(0,1,blue);(0,2,blue);(0,3,blue);(1,1,blue);(1,2,blue);(1,3,blue);(4,7,gold);(5,9,gold);(6,8,darkorange);(6,9,darkorange);(6,10,darkorange);(8,4,darkorange);(8,5,darkorange);(8,6,darkorange);(9,3,red);(10,0,darkorange);(10,1,darkorange);(10,2,darkorange) P(X2): -cells:(0,1,blue) -(0,2,blue) -(1,1,blue) -(9,3,red) +cells:(0,2,blue) +(0,4,blue) +(1,4,blue) +(10,3,red)</pre>	(empty — no world knowledge yet)
node 3 iter 4 · K 0.69 min(ID, cFD)	unchanged from node 2	new - The world contains a red cell that is moved by directional actions: ... up ... moves it 4 cells upward ... left ... moves it 1 cell left ... right ... moves it 1 cell right - The blue 2×3 block moves autonomously: each time step it shifts either left or right by 1 column. This movement is independent of the action taken - All other objects (gold cells, dark orange L-shaped groups) are stationary and never change ↑ and no gravity rule at all — refuted by the transitions above
node 16 iter 20 · K 0.79 min(ID, cFD)	<pre>inherited from node 15 — grouped by colour cells_by_colour.setdefault(colour, []).append((r, c)) ... coords_str = " ".join(f"{r},{c}" for r, c in positions_sorted) summary_parts.append(f"{colour}: {coords_str}") P(X1) = grid 12x12; bg:white; blue: 0,1 0,2 0,3 1,1 1,2 1,3 darkorange: 6,8 6,9 6,10 8,4 8,5 8,6 10,0 10,1 10,2 gold: 4,7 5,9 red: 9,3 P(X2): -blue:0,1 -blue:1,1 -red:9,3 +blue:0,4 +blue:1,4 +red:10,3</pre>	updated - The red cell moves downward by 1 cell on each noop step, unless it is at row 11 or the cell directly below is occupied by any other object - When a red cell moves ... onto a cell containing a gold cell, the gold cell disappears immediately. The red cell does not appear in that cell until the next time step - It moves right until its rightmost column reaches 11, then moves left until its leftmost column reaches 0 ... oscillating continuously